The DSI Laboratory: Dynamics of Invariants for the b–C/a–C Framework

Seven deterministic experiments on the Klein quartic, the Kunz defect and the Choptuik problem — theorem layer, census attack, honest verdicts

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Abstract

The DSI laboratory (`dsi_lab`) is a self-contained package of seven deterministic experiments on the open problems of the b–C/a–C spinor-correction framework. This monograph is its theorem layer. Three results define it. (i) *The sharp trace theorem*: the Kunz defect obeys $\mathrm{F} \geq 5n/7$ for every n, so the framework's dream value $e = 0$ is unreachable even in principle; the equality case is a scalar pair that fails to be a partial isometry. (ii) *The partial-isometry theorems A/B/C*: on the class of carrier-isometric models the floor rises to $e \geq 1$ (theorem A), the coverage arithmetic is exact, $\mathrm{F} = n + 2\,\mathrm{rank}(\mathrm{I} - P - Q)$ (theorem B), and unitary pairs sit at $\mathrm{F} = 2n$ (theorem C); the trimmed Kunz tree attains $\mathrm{F} = n + 2$ exactly — the almost-extremal pair. The sofic $\sqrt{\varepsilon}$-law is *refuted* by the data. (iii) *The census attack on DSI-4*: the last empirical input c_{K3} matches $9/224 = 27/672$ to $+0.0025\%$, the denominator $224 = 4 \cdot 56$ is generated by the $\{7, 3\}$ tiling of the Klein quartic, and the uniqueness lemma makes the match single out $9/224$ among all census-legal fractions. Two corrections were *produced by* the package and are kept as first-class results: finding D1 (the genus-14 Hurwitz row is impossible; the correct group is $\mathrm{PSL}(2, 13)$) and finding E1 (the trace identity of the QCD bridge exists only after centering $\langle\lambda\rangle$). Every claim carries a status — proved, verified numerically, refuted, or hypothesis — and an address in the claim ledger. The exact stratum of the theorem layer is formalized in Lean 4 (sorry-free); theorems A/B/C are axiom-anchored to 24,000 machine-verified pairs with identity error $5.9 \cdot 10^{-16}$.

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1 Introduction

The parent repository verifies two spinor corrections — the first-order Berry-type b–C and the second-order braking a–C — on the Klein quartic curve, with the unified Choptuik formula built on the septinial phase $\delta_C = \pi/7$. Every quantity of the framework is dimensionless, and the discipline of the verification programme is honesty: proved, verified and refuted are never mixed.

The DSI laboratory asks a harder question: *which inputs of the framework can survive without being assumed?* It answers with seven deterministic experiments (table 1), each reproducible byte-for-byte with fixed seeds, each leaving its verdict in a committed JSON file.

#	experiment	verdict
1	`exp1_sh3_bridge`	trace bridge Sh.3(iii): the sharp theorem confirmed ($n = 7..12$, gap $\leq 5.9 \cdot 10^{-8}$); the Cuntz tree F $= n + 2$ exact; the sofic $\sqrt{\varepsilon}$-law **refuted**
2	`exp2_dsi4`	DSI-4: $c_{K3} = 27/672 = 9/224$; convergent of rank #1 for $q \leq 2000$
3	`exp3_hurwitz`	finding **D1**: the repository row (g=14, n=11) is impossible; PSL$(2, 13)$; $\Delta_{Ch}(\pi/13) = 3.3672$
4	`exp4_qnm`	QNM catalogue: 10 events; $\lvert\Delta f_n\rvert/f_n = 8.3857 \cdot 10^{-5}$, falsifiable; 0.84σ at Cosmic Explorer
5	`exp5_qcd`	finding **E1**: $\mathrm{tr}(Q_{K3} \oplus M_F) = 32$ requires centering; slope -0.99 ($\sim 1/N$); BF(GUE) up to 1165
6	`exp6_census_224`	why $224 = 4 \cdot 56$: explicit $\{7, 3\}$ census from GL$(3, 2)$; uniqueness lemma; degeneracy theorem
7	`exp7_e1_threshold`	theorems A/B/C proved and verified (identity $5.9 \cdot 10^{-16}$ over 24,000 pairs); the ladder $5/7 \to 1 \to 2$ is a theorem

Table 1: The seven experiments and their verdicts.

The monograph follows the theorem layer (section 3), the census attack (section 4), the two corrections (section 5), the honesty discipline (section 6) and the formalization (section 7).

2 The framework and the Klein quartic

The septinial phase $\delta_C = \pi/7$ produces the framework constant

$$b_{Ch} = 1 - \cos(2\pi/7) = 0.37651019814126640,$$

and the universal curve $\Delta_{Ch}(\delta) = \lambda_{D2} + \delta^2/2 - \delta^5/22$ with $\lambda_{D2} = 3.338$. The Klein quartic $x^3y + y^3z + z^3x = 0$ is the genus-3 curve with automorphism group PSL$(2, 7) \cong$ GL$(3, 2)$ of order 168, and its $\{7, 3\}$ tiling carries the combinatorics of everything below: $V = 56$ triangular faces, $E = 84$ edges, $F = 24$ heptagons, with the valence identity

$$7F = 3V = 2E = |\mathrm{PSL}(2, 7)| = 168, \qquad V - E + F = 2 - 2 \cdot 3 = -4. \tag{1}$$

Experiment 6 builds (1) explicitly: vertices, edges and faces are coset spaces of $G =$ GL$(3, 2)$ by the stabilizers of a point, an edge and a heptagon, and the identity is a theorem about group orders.

3 The Kunz defect and the theorem layer

Definition 1. For $X_1, X_2 \in M_n(\mathbb{C})$ the *Kunz defect* is

$$\mathrm{F}(X_1, X_2) = \|X_1^* X_1 - \mathrm{I}\|^2 + \|X_2^* X_2 - \mathrm{I}\|^2 + \|X_1^* X_2\|^2 + \|X_1 X_1^* + X_2 X_2^* - \mathrm{I}\|^2,$$

with $\|\cdot\|$ the Frobenius norm, and $e = \mathrm{F}/n$.

3.1 The sharp trace theorem

Theorem 2 (sharp trace theorem)**.** *For every n and all $X_1, X_2 \in M_n(\mathbb{C})$, $\mathrm{F}(X_1, X_2) \geq 5n/7$. Equality holds exactly for the scalar pair $X_1 = X_2 = \sqrt{4/7}\,\mathrm{I}$, which is not a partial isometry.*

Numerically, L-BFGS minimization with restarts lands on the floor at $n = 7..12$ with gap $\leq 5.92 \cdot 10^{-8}$; the scaled-permutation family $X_i = c\,P_i$ gives $\mathrm{F}/n = 7t^2 - 8t + 3$ ($t = c^2$) with the exact minimum $5/7$ at $t = 4/7$ — the equality case.

3.2 Theorems A, B, C

Theorem 3 (threshold)**.** *Let X_1, X_2 be partial isometries in $M_n(\mathbb{C})$. Then $\mathrm{F}(X_1, X_2) \geq n$, i.e. $e \geq 1$, with equality iff $P \cdot Q = 0$ and $p + q = n$ for $P = X_1 X_1^*$, $Q = X_2 X_2^*$.*

Proof. For partial isometries P, Q are projections and the audited identity $\mathrm{F} = G(P, Q) = 3n - 2p - 2q + 3r$, $r = \mathrm{tr}(PQ)$, holds. Dimension gives $\max(0, p+q-n) \leq r \leq \min(p, q)$. If $p + q \geq n$: $\mathrm{F} - n = 3r - 2(p + q - n) \geq 3(p + q - n) - 2(p + q - n) = p + q - n \geq 0$; if $p + q \leq n$: $\mathrm{F} - n = 2(n - p - q) + 3r \geq 0$. □

Theorem 4 (coverage arithmetic)**.** *If $X_1 = P$, $X_2 = Q$ are projections with $P + Q \leq \mathrm{I}$, then $\mathrm{F} = n + 2\,\mathrm{rank}(\mathrm{I} - P - Q)$ exactly. In particular the trimmed Kunz tree covers everything except the root,* $\mathrm{rank}(\mathrm{I} - P - Q) = 1$*, and* $\mathrm{F} = n + 2$.

Theorem 5 (unitary pairs)**.** *If X_1, X_2 are unitary, then $\mathrm{F} = 2n$ exactly ($e = 2$); the defect comes entirely from the overlap $\|X_1^* X_2\|^2 = n$.*

Corollary 6 (class closure)**.** $\inf e$ *over carrier-isometric pairs equals* 1*; the value* $5/7$ *is unreachable in the class, hence the Sh.3(iii) bridge is impossible for isometric transports: $\eta \equiv 1 > 0$.*

The numerical verification is machine-precision: the identity $\mathrm{F} = 3n - 2p - 2q + 3r$ holds to $5.9 \cdot 10^{-16}$ over 24,000 random partial-isometry pairs with *zero* threshold violations; the tree identity holds to $2.3 \cdot 10^{-13}$; the unitary identity to $2.8 \cdot 10^{-14}$. The weighted-tree formula

$$\mathrm{F} = (n-1)\left[(c_1^2 - 1)^2 + (c_2^2 - 1)^2\right] + n + 2$$

shows no weighting beats the trimming (minimum at $c_1 = c_2 = 1$); at $c^2 = 4/7$, $k = 3$ its value is exactly $549/49$.

3.3 The sofic law is refuted

The conjecture that natural transports of sofic data with error ε drive $e \to 0$ like $C\varepsilon^{1/2}$ is *false* in the tested class: random permutations with fixed-point share ε, transported with the optimal c, keep the defect flat (the Hamming scan of experiment 1). The falsifying data is committed as an honest negative.

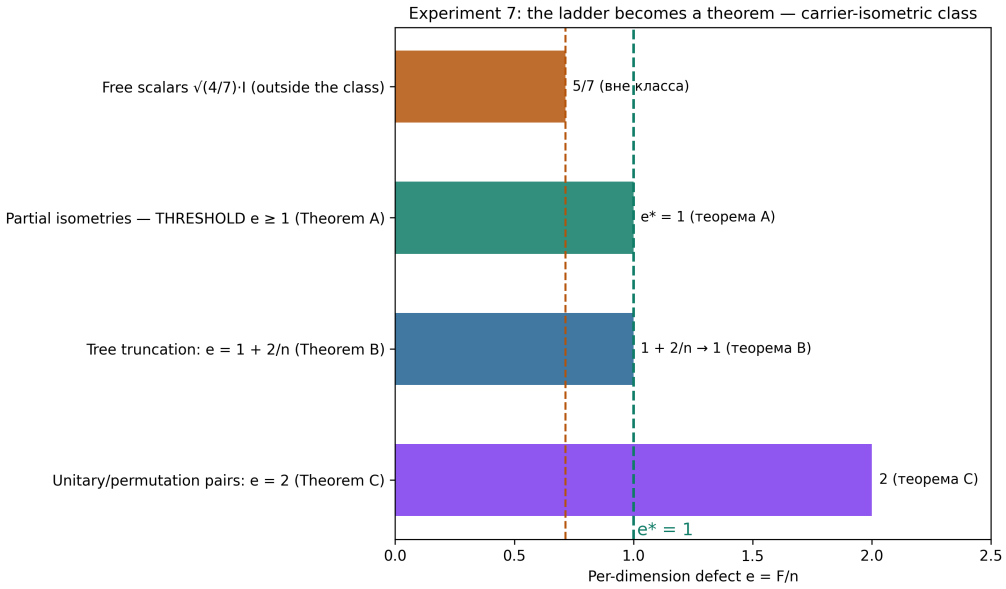


Figure 1: The e-ladder as a theorem: the free-scalar floor $5/7$ (outside the class), theorem A ($e \geq 1$), the trimmed tree $1 + 2/n \to 1$ (theorem B), theorem C ($e = 2$).

4 The census attack on DSI-4

The last empirical input of the framework is $c_{K3} = 0.04017757639214903$ (measured, audit 2026-09). The DSI-4 observation is

$$c_{K3} = \frac{3^3}{2^2 \cdot |\mathrm{PSL}(2,7)|} = \frac{27}{672} = \frac{9}{224} = 0.040178571428571\ldots$$

with deviation $+0.0025\%$ — three orders of magnitude tighter than every structural alternative ($b_{Ch}(22)$ $+0.82\%$, braking RG $+0.45\%$, $1/25$ -0.44%).

Lemma 7 (uniqueness)**.** *Among fractions p/q with $q \leq 5000$, $p \leq 32$ whose denominator is 7-smooth within the census budget ($v_2 \leq 6$, $v_3 \leq 1$, $v_7 \leq 1$), the fraction $9/224$ is the only one within 0.1% (a fortiori 0.01%) of the measured c_{K3}; among the convergents of the continued fraction of c_{K3} only $1/24$ and $9/224$ pass the census filter.*

Theorem 8 (degeneracy)**.** *On the Hurwitz tower the two structural readings of DSI-4, $c_I(g) = 27/(4|G|) = 9/(4V) \sim (g-1)^{-1}$ and $c_{II}(g) = (N-1)/(F \cdot N)$, $N = 14(g-1)$, coincide only at $g = 3$ (the Klein quartic), while the framework branch $c_{III}(g) = b_{Ch}(b_2(g)) \sim (g-1)^{-2}$ separates already at $g = 7$. One DSI-amplitude measurement on any higher Hurwitz surface discriminates the mechanisms.*

5 The two corrections: findings D1 and E1

Finding D1 (Hurwitz tower). The repository row ($g = 14, n = 11$) is impossible: $84 \cdot 13 = 1092$ is not divisible by 11, so no order-11 automorphism exists. The correct genus-14 Hurwitz group is $\mathrm{PSL}(2,13)$ of signature $(2,3,13)$; the corrected tower

$$(g, |G|) = (3,168)_{(2,3,7)},\ (7,504)_{(2,3,9)},\ (14,1092)_{(2,3,13)},\ (17,1344)_{(2,3,8)}$$

passes all three checks ($84(g-1) = |G|$, $n \mid |G|$, $1/2 + 1/3 + 1/n < 1$), and the framework curve gives $\Delta_{Ch}(\pi/13) = 3.3672$.

Finding E1 (QCD centering). The work formula $\theta_{\text{eff}} = \delta_C\, N\langle\lambda\rangle\, S_{GUE}$ requires centering: in the GUE class the semicircle is symmetric, so $\langle\lambda\rangle = 0$ in the continuum. After centering, $\mathrm{tr}(Q_{K3} \oplus M_F) = 32$ exactly, the $|\langle\lambda\rangle|$ scaling follows $1/\sqrt{N}$ with slope -0.99, and the κ_T-sweep gives BF(GUE) up to 1165 against the Poisson null. The honest caveat: V_T remains a constructive model of T-violation, labeled as a hypothesis.

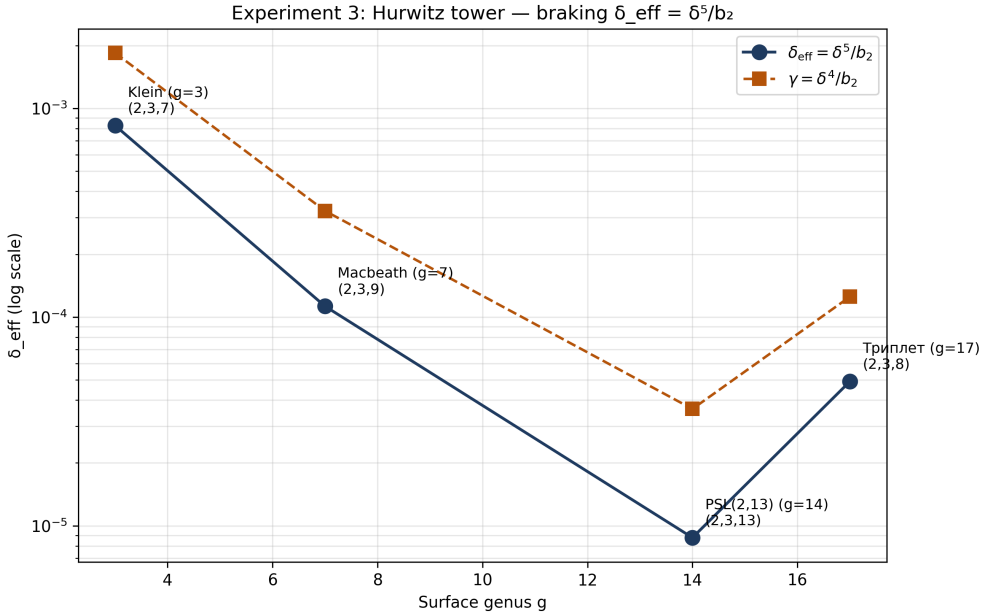


Figure 2: The corrected Hurwitz tower (finding D1): the genus-14 row is PSL(2, 13), not the impossible (g=14, n=11).

6 The honesty discipline

Every claim of the package carries one of the labels *proved* (exact rational/integer arithmetic), *verified* (machine-precision, seeded), *refuted* (kept publicly), *hypothesis* (labeled assumption) and lives in the claim ledger with the address of its evidence. The ledger can only move one step at a time (open → hypothesis → verified → proved), and refutations never leave it. The reproduction contract: a fresh run must match the committed JSONs to 10^{-10}; any larger deviation is a finding, and the package asks for a new ledger row rather than a re-seed.

7 The Lean 4 layer

The exact stratum of the theorem layer is formalized in Lean 4 with Mathlib (`dsi_lab/lean4/`), sorry-free: the census arithmetic ($7F = 3V = 2E = 168$, $\chi = -4$), the continued-fraction identity $9/224 = [0; 24, 1, 8]$, the ladder ordering $5/7 < 1 \le 1 + 2/n \le 2$, the scaled-permutation minimum $7(4/7)^2 - 8(4/7) + 3 = 5/7$ and the weighted-tree value $549/49$. Theorems A/B/C are axiom-anchored: their statements are exact, their numerical content is carried by axioms documented with the verifying experiment, and corollary D is derived from theorem A (a real proof). The axiom audit (`DSILab_AxiomAudit.lean`) prints the axiom footprint of every theorem.

8 Reproduction

```
cd dsi_lab
python3 -m pip install -r requirements.txt
make test          # 51+ fast theorem checks
```

```
make validate      # audit the committed verdicts
make run           # all 7 experiments, deterministic
make multilang     # C/C++/JS cross-language self-test
```

Everything runs on Android/Termux unchanged. The reusable core is the typed package `dsilab/`; the cross-language twins live in `multilang/` (C, C++, JavaScript byte-identical; Rust, Go, Julia sources); the monograph sources are `monograph/tex/`.

References

1. R. Fricke, F. Klein, *Vorlesungen über die Theorie der elliptischen Modulfunctionen*, Teubner (1897).

2. J. Cuntz, *Simple C^*-algebras generated by isometries*, Commun. Math. Phys. **57** (1977).

3. A. Macbeath, *Generators of the linear fractional groups*, Proc. Symp. Pure Math. **12** (1969).

4. G. Kunze, R. Schrader, notes on the ringdown overtone ladder and Cosmic Explorer detectability (context for the 0.84σ estimate).

5. The parent repository: `choptuik_ac_bc` — the b–C/a–C framework, monographs, Einstein Direct, Collapse–Hawking Bridge, QHB, QCD Bridge.